Executive Summary – Solar Investment Analysis

Overview

This analysis evaluates a residential solar asset using 4 years of income data, focusing on profitability, cash flow stability, and operational optimization opportunities.

The loan is expected to end in December 2034, creating several years of fully unleveraged profit after break-even.

Break-even	Cleaning ROI	Winter Risk
Sep 2029	~248%	Nov–Jan

Key Insights

Liquidity Risk: Winter months (Nov–Jan) fail to cover loan payments.
Implication: Requires active cash flow management.

Maintenance ROI: ILS 2,899 uplift vs ILS 1,170 cost (~248% ROI); ILS 1,729 net profit.
Implication: Shows strong potential and should be validated before scaling.

Efficiency Gap: August underperforms despite higher radiation.
Implication: Operational constraints reduce revenue conversion.

Technical Approach

Analysis performed using Python (Pandas) and SQL-based logic for data cleaning, aggregation, and time-series alignment. Data preprocessing included anomaly detection, duplicate invoice identification, and correction of misaligned records.

Why it matters: Cleaning and realigning the data protects financial conclusions from distorted revenue timing.

Assumptions & Limitations

Break-even assumes stable historical performance. Panel degradation, weather variability, and tariff changes were not explicitly modeled.

Final Takeaway

The investment is fundamentally strong, generating value through both cash flow and equity growth. However, its performance is not fully optimized and requires active management.

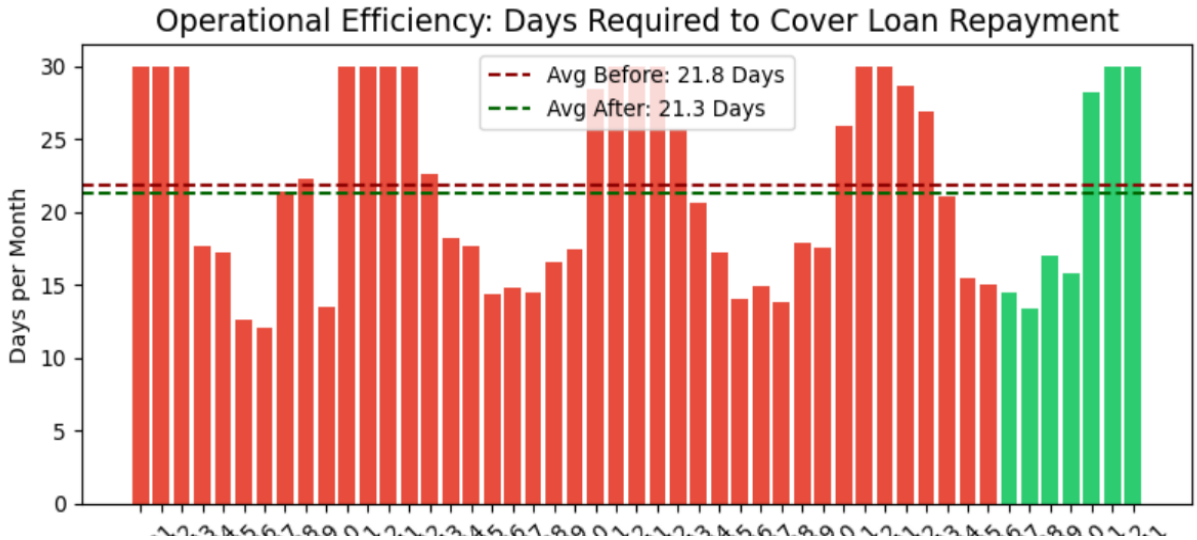
Priority 1: Address structural winter liquidity gaps to ensure stable cash flow.

Priority 2: Investigate efficiency losses during peak solar periods ("August Gap").

If managed effectively, the asset can transition into a high-yield investment as it approaches full debt repayment in 2034.

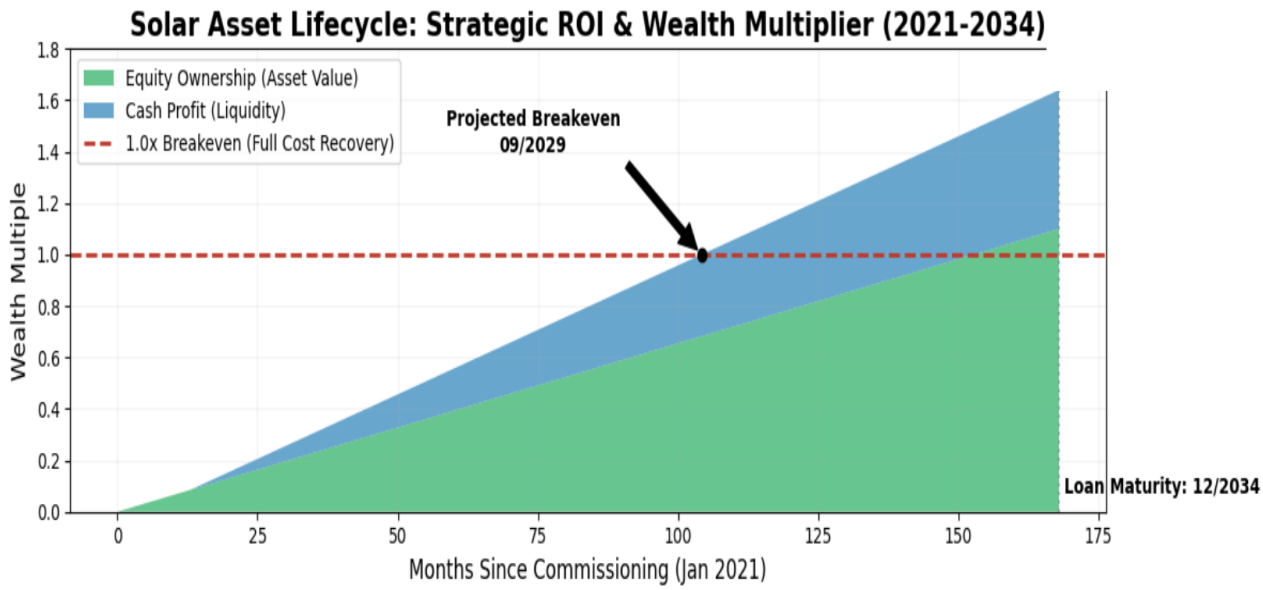
Appendix – Supporting Visual Analysis

Graph 1 – Loan Coverage Efficiency



Cleaning improves cash flow efficiency and slightly reduces time required to cover loan obligations.

Graph 2 – Asset Value Growth



The asset reaches projected break-even in September 2029, while value continues to accumulate through both cash profit and equity ownership until loan maturity in December 2034.